

SIMATS ENGINEERING

ASSESSMENT TOOL III – VISUALIZATION CRITIQUE & REDESIGN

Course: Data Handling and Visualization	Code: DSA0601	CO: CO3
Duration: 120 min	Total Marks: 100	Reg No: 192324284
Name: V.P Sneha		

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods.

(BL4) Activity Tool: Visualization Critique & Redesign Weightage: 20% **Code of Conduct:**

I V.P Sneha(192324284) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _v.p sneha

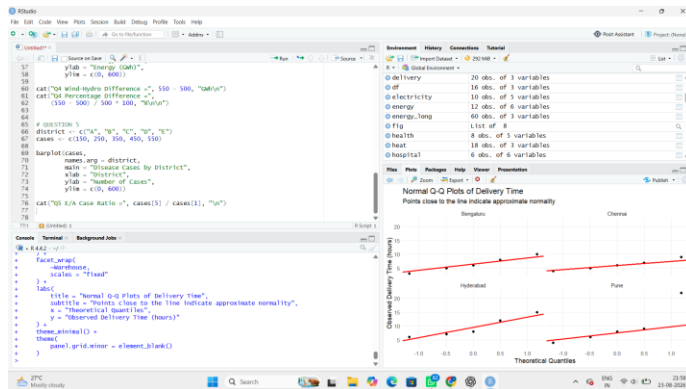
Criteria	Evaluation of Existing Visualization (25%)	Redesign Strategy (25%)	Application of Vis. Principles (20%)	Organization (15%)	Accuracy (10%)	Clarity (5%)	Total
Q1							
Q2							
Q3							
Q4							
Q5							

AT3 – Visualization Critique & Redesign

R Programming Assessment – Distribution and Proportion Visualization

Instructions: For each question, use R/RStudio to reproduce the given visualization, critique its effectiveness, identify misleading or inappropriate visual encodings, and redesign it using a more suitable visualization. Include the R code, redesigned visualization, and a concise justification of your design decisions.

Question 1 – Critique and Redesign an ECDF and Q-Q Plot



Problems in the original visualization

At least three problems are:

1. Inconsistent axis scaling makes warehouse comparisons difficult.
2. Missing Q-Q reference line makes normality assessment harder.
3. Excessive gridlines create visual clutter.
4. Labels are not sufficiently clear.
5. The extreme value of 22 hours in Pune may be difficult to identify without appropriate scaling.

Interpretation

Pune has a clear high delivery-time observation of 22 hours, suggesting a possible outlier/right-tail observation.

The Q-Q plot helps determine whether the data follow a normal distribution. Strong deviations from the reference line indicate non-normality.

Is ECDF alone sufficient?

No.

An ECDF is useful for comparing distributions and identifying unusual tail behavior, but it should be paired with a Q-Q plot, boxplot, or another diagnostic plot to investigate outliers and normality more clearly.

Question 2 – Critique and Redesign Many Distributions Using Bean Plots

The screenshot displays the RStudio interface with the following components:

- Source Editor:** Contains R code for data manipulation:


```
# 1.1.1. Summarize
# 1.1.2. Filter
# 1.1.3. Arrange
# 1.1.4. Sample
# 1.1.5. Join
# 1.1.6. Split
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# 1.4.57. Transform
# 1.4.58. Summarize
# 1.4.59. Filter
# 1.4.60. Arrange
# 1.4.61. Sample

```

The original visualization has:

- Highest concentration

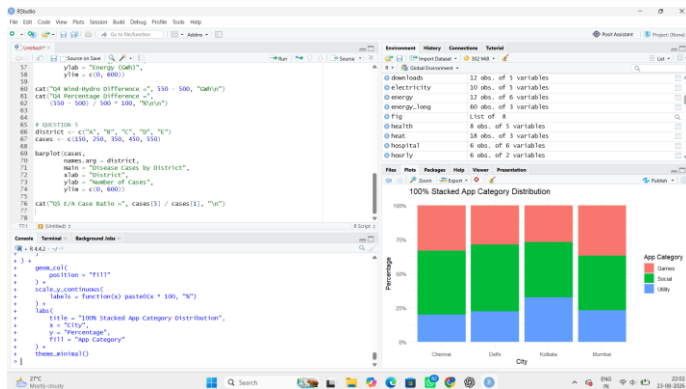
Greatest spread

- Physics = $85 - 58 = 27$
- Chemistry = $82 - 52 = 30$
- Mathematics = $90 - 61 = 29$
- Biology = $88 - 55 = 33$

Biology has the greatest range/spread = 33 marks.

A common Y-axis from 0–100 makes comparisons more honest.

Question 3 – Critique and Redesign Proportion Visualizations



Four problems with the pie charts

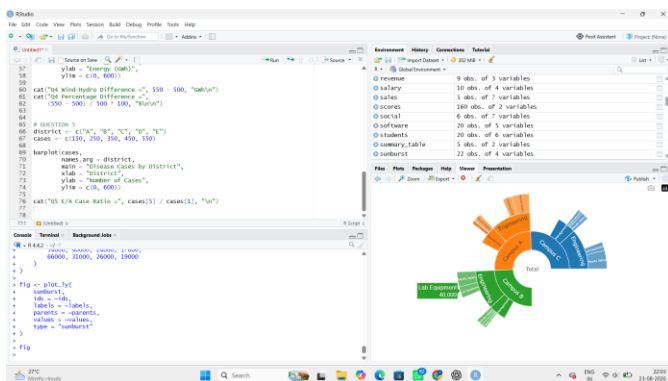
1. Exploded slices exaggerate visual importance.
2. Mismatched colors make the same category difficult to track across cities.
3. Pie slices are difficult to compare accurately.
4. Inconsistent labeling creates confusion.
5. Comparing angles and areas is less accurate than comparing bar lengths.

Better chart

For comparing city-wise proportions, the side-by-side bar chart is generally better because all categories share a common baseline.

The 100% stacked bar is better for showing the overall composition of each city.

Question 4 – Critique and Redesign Nested Proportions



Critique

Problems with the original treemap:

- Small areas have unreadable labels.
- Colors do not clearly communicate hierarchy.
- Department and expense type can be difficult to distinguish.
- Too much information is displayed at once.
- Small rectangles are difficult to compare.

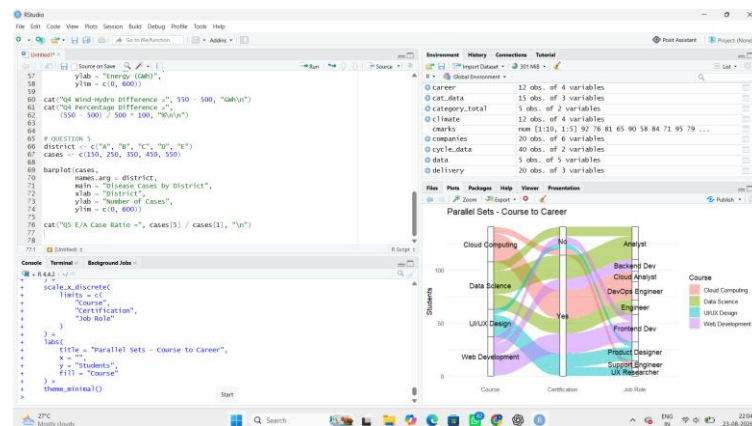
Board presentation recommendation

I recommend the redesigned treemap for a board presentation because it gives a quick view of spending magnitude and allows categories to be compared visually.

The sunburst is better when the main purpose is explaining the hierarchy:

Campus → Department → Expense Type.

Question 5 – Critique and Redesign Flow-Based Proportions



Four possible problems in the original Sankey

1. Near-identical colors make flows difficult to distinguish.
2. Unlabelled flows make the exact meaning difficult to understand.
3. Inconsistent node widths can distort perceived flow magnitude.
4. Small flows may be difficult to see.
5. Poor ordering can cause excessive crossing of links.
6. Lack of numerical labels makes exact comparison difficult.

Redesign improvements

The redesigned chart should use:

- Clear node labels.
- Meaningful colors.
- Consistent flow widths.
- Clear hierarchy.
- Course → Certification → Job Role ordering.
- Minimal unnecessary decoration.

Major career flows

The largest individual flow is:

Data Science → Yes → Analyst = 22 students

Other major flows include:

Web Development → Yes → Frontend Dev = 19

Cloud Computing → Yes → DevOps Engineer = 17

